Hamming Distance

Input: Two Strings
Output: The number of mismatched symbols

Symbol Matching Problem

Input: Two Strings
Output: The greatest number of matched symbols in any "alignment" of the two strings

LCS

0	3	5	7	7
1	4	7	11	13
4	10	17	20	24
8	14	22	21	
13		30	32	34

Manhattan Tourist Problem

Find longest path from source to sink in a DAG

We can form a bijection from alignments to paths

Origin of dp is trying to sound important

$fib(n) = fib(n-1) + fib(n-2)$
is bad if implemented naively as the amount of calls blows up exponentially.

We use dynamic programming to memoize the computed values.